

Addendum: The Casimir Effect and Number-Theoretic π

Addendum to: "The Geometric Vocabulary in Quantum Field Theory: Circumferences and Surfaces in One-Loop QED"

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A1. Introduction

The main paper identified two pathways by which π enters the one-loop Schwinger correction $\alpha/(2\pi)$: geometric π (from angular integrals over momentum-space sphere surfaces and phase circumferences) and statistical π (from Gaussian integrals via $\Gamma(1/2) = \sqrt{\pi}$), with the latter flagged as the kill switch for higher-loop extensions.

This addendum extends the analysis to the Casimir effect (Equation #50 in the 55-equation catalog) and reports three results: the identification of a third pathway for π , an explicit boundary where the geometric vocabulary ceases to apply, and a connection between the Casimir boundary and the two-loop QED correction through the shared appearance of $\zeta(3)$.

A2. Factor Decomposition of the Casimir Force

The Casimir force per unit area between parallel conducting plates separated by distance d is (Casimir 1948):

$$F/A = -\pi^2 \hbar c / (240 d^4)$$

In D spatial dimensions, for a massless scalar field with Dirichlet boundary conditions (Ambjørn and Wolfram 1983, Elizalde 1995):

$$E/A = -[\Gamma((D+1)/2) \times \zeta(D+1)] / [2^{(D+1)} \times \pi^{((D+1)/2)}] \times \hbar c / d^D$$

The electromagnetic force acquires factors of $(D-1)$ polarizations and D from differentiation:

$$F/A = -D(D-1) \times [\Gamma((D+1)/2) \times \zeta(D+1)] / [2^{(D+1)} \times \pi^{((D+1)/2)}] \times \hbar c / d^{(D+1)}$$

At $D = 3$, the scalar energy coefficient is $-\pi^2/1440$. Multiplying by 2 polarizations gives $-\pi^2/720$ (energy), and differentiating gives $-\pi^2/240$ (force).

The factor 240 decomposes uniquely as:

- **2** = electromagnetic polarization count ($D-1$ at $D = 3$). Non-geometric — a degeneracy count, same category as the 2 in Planck's law (#48).

- **3** = spatial dimensionality D (from $\partial/\partial d$ of $d^{(-D)}$). Geometric — same word as in Equations #53 and #55.
- **40** = mode-counting factors from $\zeta(D+1)$, combined with momentum-space measure.

This is the only factorization in which every factor has independent D -dependence, a distinct physical origin, and produces the correct result at arbitrary D .

Factor-by-factor identification

Factor	Origin	Geometric?	D-test
$\hbar c/d^{(D+1)}$	Quantum-relativistic energy scale	Dimensional backbone	—
D	$\partial/\partial d$ of $d^{(-D)}$	Yes (dimensionality)	Pass
$(D-1)$	Transverse polarizations	No (degeneracy count)	Pass
$\Gamma((D+1)/2)$	Angular integral over k -space sphere	Yes (surface)	Pass
$2^{(D+1)}, \pi^{((D+1)/2)}$	Momentum-space measure $d^{(D-1)}k/(2\pi)^{(D-1)}$	Yes (phase circumference)	Pass
$\zeta(D+1)$	Regularized sum over discrete modes	Mixed (see Section A3)	Pass

A3. A Third Pathway for π

The main paper identified two pathways. The Casimir effect reveals a third.

At $D = 3$, the mode-summation factor $\zeta(4) = \pi^4/90$. The π^4 does not come from angular integration (Pathway 1) or Gaussian integration (Pathway 2). It enters through Euler's 1740 formula:

$$\zeta(2k) = |B_{2k}| \times (2\pi)^{(2k)} / [2 \times (2k)!]$$

where B_{2k} is the $2k$ -th Bernoulli number. The $(2\pi)^{(2k)}$ is the $(2k)$ -th power of the period of the cotangent function, entering because $\zeta(s)$ is related to $\cot(\pi z)$ through the generating function $t/(e^t - 1)$.

The three pathways are:

Pathway 1 — Geometric π . From angular integrals over sphere surfaces and phase boundaries. Mathematically: $\int d\Omega$ over $S^{(D-1)} \rightarrow \Omega_{-}(D-1) \rightarrow \pi^{((D-1)/2)}$. Present in Coulomb's law, the Schwinger correction, and the momentum-space measure of the Casimir calculation.

Pathway 2 — Statistical π . From Gaussian integrals via $\Gamma(1/2) = \sqrt{\pi}$. Not present in the Casimir effect. Flagged in the main paper as the kill switch for higher loops.

Pathway 3 — Number-theoretic π . From Euler's formula for $\zeta(2k)$, entering through the periodicity of the Bernoulli generating function. Present in the Casimir effect (via $\zeta(4) = \pi^4/90$), the Stefan-Boltzmann law, and black-body radiation.

Classification: The identification of (2π) inside Euler's formula as a circumference (the KMS thermal circle of the Bose-Einstein distribution, which shares the generating function $t/(e^t - 1)$) is Speculative. The three-pathway classification itself is New.

Observational distinctness

The three pathways are observationally distinguishable through the D-test:

- **Pathway 1** produces π for all values of D.
- **Pathway 2** produces $\sqrt{\pi}$ whenever a Gaussian integral is performed, independent of D.
- **Pathway 3** produces π only when $\zeta(D+1)$ has an even argument (D odd). When D is even, $\zeta(D+1)$ evaluates to an irrational constant with no known closed form involving π .

The values $\zeta(3) \approx 1.202$, $\zeta(5) \approx 1.037$, $\zeta(7) \approx 1.008$ have no known expression as rational multiples of powers of π . Apéry (1978) proved $\zeta(3)$ is irrational; no π relation is known for any $\zeta(\text{odd})$.

Spatial dimensions D	ζ argument	Contains π ?
1	$\zeta(2) = \pi^2/6$	Yes (Pathway 3)
2	$\zeta(3) = 1.2021\dots$	No
3	$\zeta(4) = \pi^4/90$	Yes (Pathway 3)
4	$\zeta(5) = 1.0369\dots$	No
5	$\zeta(6) = \pi^6/945$	Yes (Pathway 3)
6	$\zeta(7) = 1.0083\dots$	No

The on/off pattern is sharp. This behavioral difference — Pathway 1 always produces π , Pathway 3 only at odd D — is the empirical proof that they are distinct pathways.

Two pathways in one equation

The Casimir force at $D = 3$ contains π from two simultaneous pathways: Pathway 1 through the momentum-space measure $(2\pi)^2$, and Pathway 3 through the $(2\pi)^4$ inside $\zeta(4)$. In standard notation both appear as " π^2 " and the two origins are invisible. The vocabulary distinguishes

them. This is the only equation in the 55-equation catalog where π enters through two different pathways simultaneously.

A4. Cross-Reference Table

Equation	Pathway 1 (Geometric)	Pathway 2 (Statistical)	Pathway 3 (Number-theoretic)
Coulomb's law (#46)	✓ (4π flux surface)	—	—
Gauss's law (#8-10)	✓ (4π flux surface)	—	—
Phase equations (#41-45)	✓ (2π phase circle)	—	—
Schwinger correction	✓ (momentum sphere, phase)	—	—
Casimir effect (#50)	✓ (momentum measure)	—	✓ ($\zeta(4)$)
Stefan-Boltzmann law	✓ (angular integration)	—	✓ ($\zeta(4)$)
Planck radiation (#48)	✓ (phase, via $h = 2\pi\hbar$)	—	✓ ($\zeta(4)$ in integrated form)
Thermal de Broglie (#51)	✓ (phase circumference)	?	—
Higher-loop QED	✓ (momentum sphere)	✓ ($\Gamma(1/2) = \sqrt{\pi}$)	?

A5. The Even/Odd Boundary

The D-dimensional Casimir effect reveals a boundary where the vocabulary stops.

At odd D ($D = 1, 3, 5, \dots$), the Casimir force coefficient decomposes fully into geometric factors. The scalar energy coefficient is a rational multiple of $\pi^{((D+1)/2)}$, with the rational part controlled by Bernoulli numbers:

$$C_D(\text{scalar, odd } D) = -\Gamma((D+1)/2) \times |B_(D+1)| / [2 \times (D+1)!] \times \pi^{((D+1)/2)}$$

D	$C_D / \pi^{((D+1)/2)}$
1	$-1/24$
3	$-1/1440$
5	$-1/30240$
7	$-1/403200$

At even D ($D = 2, 4, 6, \dots$), the coefficient contains $\zeta(\text{odd})$ — irrational, with no known expression involving π and no known geometric reading.

The mathematical fact that odd-D Casimir coefficients are expressible through Bernoulli numbers while even-D coefficients involve $\zeta(\text{odd})$ is established in the literature (Cheng et al. 2010, Dolan and Nash 1992). The connection to heat kernel expansions and the Atiyah-Singer index theorem through the same Bernoulli numbers is reviewed in Vassilevich (2003). The present contribution is the vocabulary reading: this split corresponds precisely to the boundary between equations whose dimensionless factors admit geometric interpretation and those that do not.

A6. $\zeta(3)$ Across Contexts

The boundary identified in Section A5 is not specific to the Casimir effect. The same $\zeta(3)$ that resists geometric reading in the even-D Casimir coefficient also appears analytically in the two-loop QED correction to the electron $g-2$ (Petermann 1957, Sommerfield 1958), where the coefficient C_2 contains the term $3\zeta(3)/4$. In that context, $\zeta(3)$ enters through dilogarithm integrals over Feynman parameters: $\int_0^1 \text{Li}_2(x)/x \, dx = \zeta(3)$.

The vocabulary hits the same wall in both cases. In the Casimir effect, $\zeta(3)$ appears because the mode sum over discrete vacuum fluctuations at even D evaluates to $\zeta(\text{odd})$. In two-loop QED, $\zeta(3)$ appears because the nested integral over virtual particle momenta evaluates to the same class of function. Neither admits a decomposition into circumferences, surfaces, or bridges.

This suggests that the limitation is general: whenever $\zeta(\text{odd})$ enters a physical quantity — whether through boundary-condition mode sums (Casimir) or loop-momentum integrals (QED) — the geometric vocabulary encounters factors with no known geometric reading. The even/odd ζ boundary is a property of the vocabulary's reach, not of any particular physical system.

A7. Updated Kill Switches

The main paper identified one kill switch ($\Gamma(1/2) = \sqrt{\pi}$ at higher loops). The Casimir analysis adds a second.

Kill switch 1 (unchanged): Statistical π at higher loops. If higher-loop QED corrections introduce $\Gamma(1/2) = \sqrt{\pi}$ in a way that resists geometric interpretation, Pathway 2 becomes a genuine obstacle. In dimensional regularization at $D = 4 - 2\varepsilon$, Type 1 factors scale as $\pi^{(2-\varepsilon)}$ (tracking dimension) while Type 2 factors from Feynman parameter integrals such as $B(1/2, 1/2) = \pi$ are ε -independent (not tracking dimension). This ε -dependence provides an operational method, in principle, for separating the two pathways in explicit calculations.

Kill switch 2 (new): Closed form for $\zeta(\text{odd})$ involving π . If a closed form for $\zeta(3)$, $\zeta(5)$, or any $\zeta(\text{odd})$ is discovered as a rational multiple of a power of π , the behavioral distinction between Pathways 1 and 3 collapses, and the three-pathway classification reduces to two. No such closed form is currently known.

A8. Summary of Addendum

1. The factor 240 in the Casimir force decomposes uniquely as $2_{\text{pol}} \times 3_{\text{dim}} \times (\zeta \text{ factors})$, verified at arbitrary D .
 2. π enters the Casimir force through two simultaneous pathways — geometric (momentum-space measure) and number-theoretic (Euler's formula for $\zeta(4)$). This is the only two-pathway equation in the catalog.
 3. A third pathway for π (number-theoretic, from $\zeta(2k)$) is observationally distinct from the geometric pathway: it produces π only at odd D and vanishes at even D .
 4. The vocabulary stops at even D , where $\zeta(\text{odd})$ has no known geometric reading. The same $\zeta(3)$ appears in both the even- D Casimir effect and the two-loop QED correction, indicating a general boundary of the vocabulary rather than a system-specific limitation.
 5. The ε -dependence in dimensional regularization provides an operational method for distinguishing Type 1 (geometric) from Type 2 (statistical) π contributions at higher loop orders.
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